INTERMEDIATE HTML

HTML Tables

HTML tables are used to display data in rows & columns, similar to Excel sheets.

* Basic Table Structure

```
<table>
  <tr>
    <td> Data </td>
    <td> Data </td>
  </tr>
</table>
```

* Tags Explained

Tag	Meaning	Purpose
<table>	table container	Wraps the entire table
<tr>	Table Row	Creates a row
<td>	Table Data	Cell inside a row
<th>	Table Header	Bold + centered heading cell

* Proper Semantic Table Structure

```
<table>
  <thead>
    <tr>
      <th> Name </th>
      <th> Age </th>
```

```
        </tr>
    </thead>

    <tbody>
        <tr>
            <td> Aradhya <td>
            <td> 18 </td>
        </tr>
    </tbody>

    <tfoot>
        <tr>
            <td> Total </td>
            <td> 1 </td>
        </tr>
    </tfoot>
</table>
```

* Why use <thead>, <tbody>, <tfoot>?

- Improves readability
- Helps screen readers (accessibility)
- Makes Tables easier to style with CSS
- Helps browser understand table structure

- Table Attributes (Old vs Modern)

Old HTML (avoid)

```
<table border = '1'>
```

Modern Way (CSS handling styling)

```
table {
    border-collapse : collapse;
}
```

```
td, th {
    border: 1px solid black;
}
```

* Common Table Mistakes
- Putting <td> directly inside <table>
- Forgetting <tr>
- Using tables for page layout (DON'T)

HTML FORMS

Forms allow users to send data to a server.

* Basic Form Structure

```
<form>
    <input type="text">
</from>
```

But this is incomplete

* Complete Form Structure

```
<form action="submit.html" method="post">
    <label> Name : </label>
    <input type="text" name="username">
    <input type="submit" >
</form>
```

FORM ATTRIBUTES

- action

```
<form action="submit.html">
```

→ Defines where the data goes.

Examples
- action = "index.html"
- action = "mailto:abc@gmail.com"
- action = "server.php"

- method

  < form method = "post >

Method	Use
get	Appends data in URL
post	Sends data Securely

→ Example :

< form action = "submit.html" method = "get">

URL becomes

submit.html? name = Aradhy & age = 20

Post is preferred for sensitive data

INPUT Types:

* Text Input

<input type = "text">

Used for :
- Names
- Titles
- Short text

* Email Input

<input type = "email">

Automatically checks for valid email format

- Password Input

  <input type="password">

Hides typed characters.

- Submit Button

  <input type="submit">

Sends form data

- Radio Buttons

  <input type="radio" name="gender" value="male
  <input type="radio" name="gender" value="female

Same name = only one option selectable

- Checkboxes

  <input type="checkbox">

Multiple Selections allowed

- Data Input

  <input type="date">

opens calendar UI

- File Input

  <input type="file">

Upload files

- Range Slider

  <input type="range" min="0" max="10">

- Colour Picker

```
<input type="Color">
```

Labels

- WRONG way

```
<label> Name </label>
<input type="text">
```

- CORRECT way

```
<label for="name">Name</label>
<input type="text" id="name">
```

- Clicking label focuses input
- Helps screen readers
- Industry standard

PLACEHOLDER vs VALUE

```
<input type="text" placeholder="Enter name">
```

Attribute	Meaning
placeholder	Hint text
value	Default actual value

Required & Validation

```
<input type="email" required>
```

- Browser prevents form submission
- No JavaScript needed

TEXTAREA (Multi-line Input)

<textarea rows="5" cols="30"></textarea>

Used for
- Messages
- Feedback
- Long Descriptions

BUTTON vs INPUT SUBMIT

<button> Submit </button>
vs
<input type="submit">

- <button> can contain HTML
- <input> cannot

FORM + MAILTO (Angela Yu Eg)

<form action="mailto: test@gmail.com" method="post" enctype="text/plain">

- Opens email client
- Not reliable for production
- Only for learning/demo

Div & Span (Structural Tags)

-
  - Block element
  - Used to group content
  - Takes full width

```
<div>
    <h1> Hello </h1>
</div>
```

-
  - Inline element
  - Used inside text

```
<p> This is <span> important </span> </p>
```

HTML ENTITIES

Used to reserved characters

Entity	Output
<	<
>	>
	Space
©	©

HTML COMMENTS

```
<!-- This is a comment -->
```

- Ignored by browser
- Used for notes / debugging

HTML SEMANTIC MEANING (Pre CSS foundation)

HTML is not for design, it's for structure & meaning

Bad:

```
<b> Important </b>
```

Good:

```
<strong> Important </strong>
```

Why?
- Screen readers
- SEO
- Accessibility

Most important Common Mistakes

- Using tables for layout
- Forgetting [name] attribute in inputs
- No Labels
- No form method
- Styling in HTML instead of CSS

FINAL SUMMARY

After mastering Intermediate HTML you should:

- Understand how data flows from browser to server
- Build real forms
- Structure clean, semantic HTML
- Write accessible & professional markup
- Be ready for CSS & Java Script.